

Face De-Identification using Neural Radiance Fields

Maj Bregar
IBB 2026/27, FRI, UL
mb95331@student.uni-lj.si

Abstract—This work investigates face de-identification using Neural Radiance Fields and latent-space editing. A balanced subset of 1000 images was constructed from the MORPH-II dataset, and each image was projected into the EG3D W^+ latent space. Three de-identification techniques were evaluated: Middle Average Interpolation, Random Offset, and Image-Matched Residual editing. Their effects were measured using image-quality metrics, age and sex prediction, and AdaFace identity verification. Increasing identity truncation consistently improved de-identification but reduced visual similarity and demographic attribute preservation. Image-Matched Residual editing achieved the strongest identity suppression, reaching a de-identification success rate of 97.29% and an AUC of 0.8466 at maximum truncation, but also produced the greatest degradation in image quality and age preservation. The results demonstrate that NeRF latent editing can provide effective face de-identification, while highlighting the need for semantically controlled editing methods that better preserve biometric attributes.

I. INTRODUCTION

Advances in face-recognition systems have increased concerns regarding the collection, storage, and automated analysis of facial biometric data. Face de-identification aims to reduce these privacy risks by altering identity-related facial information so that recognition systems can no longer reliably associate an image with the original subject. At the same time, useful visual characteristics should be preserved whenever possible.

Although many face de-identification methods have been proposed, comparatively little work has explored Neural Radiance Fields (NeRFs) for this purpose. NeRF-based face models are particularly relevant because they support three-dimensional reconstruction and viewpoint-consistent rendering. These properties make it possible to modify identity-related appearance before rendering the subject from arbitrary camera poses.

This report investigates the use of NeRF-based facial reconstruction for face de-identification. It describes the construction of an evaluation dataset, presents three simple latent-embedding editing techniques, and evaluates their effects on image quality, age and sex preservation, and identity-recognition performance.

II. RELATED WORK

Several existing works have explored face de-identification and neural face reconstruction using Neural Radiance Fields.

In the article *Privacy-Enhancing Face Biometrics: A Comprehensive Survey* [1] Meden *et al.* provide a comprehensive overview of biometric privacy-enhancing techniques, including evaluation methods, datasets, standards, and regulatory considerations, thereby establishing the broader context for face de-identification research.

Meden *et al.* propose k-Same-Net [2], a GNN-based face de-identification method that generates surrogate faces within a k -anonymity framework, providing privacy guarantees for a closed set of identities while allowing semantic control over attributes such as facial expression, age, and sex. This approach is particularly relevant because it demonstrates attribute-aware de-identification, which is not explicitly supported by the latent-editing techniques evaluated in this work.

Zhang *et al.* present FaceDNeRF [3], a generative NeRF framework that reconstructs high-quality 3D faces from a single image and supports semantic editing and relighting through 3D GAN inversion and latent-diffusion guidance. Although FaceDNeRF was not developed specifically for de-identification, its face reconstruction forms the technical foundation of our de-identification pipeline.

III. METHODOLOGY

A. Age-Sex Dataset Generation

The experiments were based on the MORPH-II dataset [4], using a preprocessed distribution obtained from Kaggle [5]. The dataset contains frontal facial images acquired largely from law-enforcement records, with each image annotated using the subject's age and sex. Owing to the data-acquisition process, the original training set exhibits substantial demographic imbalance. Male subjects account for 86.06% of the images, and most subjects are younger than 50 years, although the full age range extends from 16 to 77 years. The age distribution of the original training set is shown in Figure 1.

To reduce these imbalances, a subset of 1000 images was constructed for the de-identification experiments. The subset contains an equal number of male and female subjects, resulting in a 50/50 sex distribution. Images were sampled to obtain an approximately uniform age distribution, with similar numbers of samples for each represented age. Less frequent ages, including subjects older than 60 years, were retained whenever sufficient samples were available. This sampling strategy does not eliminate

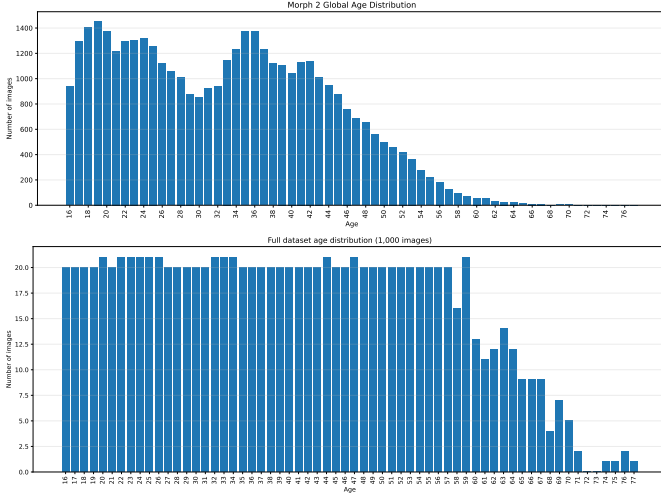


Fig. 1. Age distributions of the original MORPH-II training set (top) and the sampled age-sex dataset (bottom).

all demographic bias in MORPH-II, but it produces a more balanced evaluation set for assessing face de-identification performance across age and sex groups.

The subset size was limited primarily by the computational cost of processing individual images with the NeRF-based de-identification pipeline.

B. De-Identification Pipeline

A full schema of the de-identification pipeline is shown in Figure 2.

The raw dataset images are first preprocessed to obtain aligned facial images and corresponding camera parameters. MTCNN [6] detects five facial landmarks used for alignment. Deep3DFaceRecon [7] estimates the 3D facial parameters and head pose needed for camera estimation. The images are then aligned, cropped, and resized to a consistent resolution. Finally, the estimated pose is converted into camera parameters and adjusted to match the cropped images and the camera convention used by EG3D.

Each preprocessed image is then inverted into the EG3D W^+ latent space using its estimated camera pose. During inversion, the generator remains fixed while the latent embedding is optimized to reconstruct the input image.

The resulting embedding is used as the pivot for Pivotal Tuning Inversion (PTI), which fine-tunes the generator weights to further improve reconstruction quality. The final W^+ embedding and fine-tuned generator are then used to render the subject from custom camera poses.

C. De-Identification Using Latent Space Embedding Editing

Face de-identification is achieved by altering the image latent space embedding before rendering. We present

3 simple techniques which alter the appearance of the rendered image:

- **Random Offset.** A random tensor r is generated with the same shape as the source W^+ embedding w . Its component along $w - w_{\text{avg}}$ is removed to avoid reproducing standard average-latent truncation:

$$r_{\perp} = r - \frac{\langle r, w - w_{\text{avg}} \rangle}{\|w - w_{\text{avg}}\|^2} (w - w_{\text{avg}}).$$

The remaining direction is normalized and scaled using the typical latent-space radius R multiplied by a user-defined coefficient α :

$$\Delta = \alpha R \frac{r_{\perp}}{\|r_{\perp}\|}, \quad \alpha = 2.0.$$

Here, R is the median distance of randomly sampled latent vectors from the mean of the sampled latent bank. The final embedding is obtained using truncation strength t :

$$w_{\text{edited}} = w + t\Delta.$$

This method serves as a random de-identification baseline.

- **Middle Average Interpolation.** The third to seventh style layers of the source W^+ embedding are interpolated toward the corresponding layers of the generator’s average embedding:

$$w_{\text{edited}}^{(l)} = w^{(l)} + t \left(w_{\text{avg}}^{(l)} - w^{(l)} \right), \quad l \in \{3, \dots, 7\}.$$

All remaining style layers are left unchanged. The method moves selected facial features toward the generator’s average appearance while preserving information encoded in the unmodified style vectors.

- **Image-Matched Residual.** A bank of donor embeddings is sampled from the generator’s mapping network and rendered using the same camera pose as the source image. Each donor embedding is scored to favour both a large latent-space displacement and good visual agreement with the source. The donor with the highest score is selected.

Rather than interpolating directly toward the full donor W^+ , the method first computes the mean style vector of the source and donor embedding layers:

$$\bar{w}_s = \frac{1}{L} \sum_{l=1}^L w_s^{(l)}, \quad \bar{w}_d = \frac{1}{L} \sum_{l=1}^L w_d^{(l)}.$$

The shared latent shift $(\bar{w}_d - \bar{w}_s)$ is then applied equally to every source style layer:

$$w_{\text{edited}}^{(l)} = w_s^{(l)} + t(\bar{w}_d - \bar{w}_s).$$

This moves the source embedding toward a visually compatible donor.

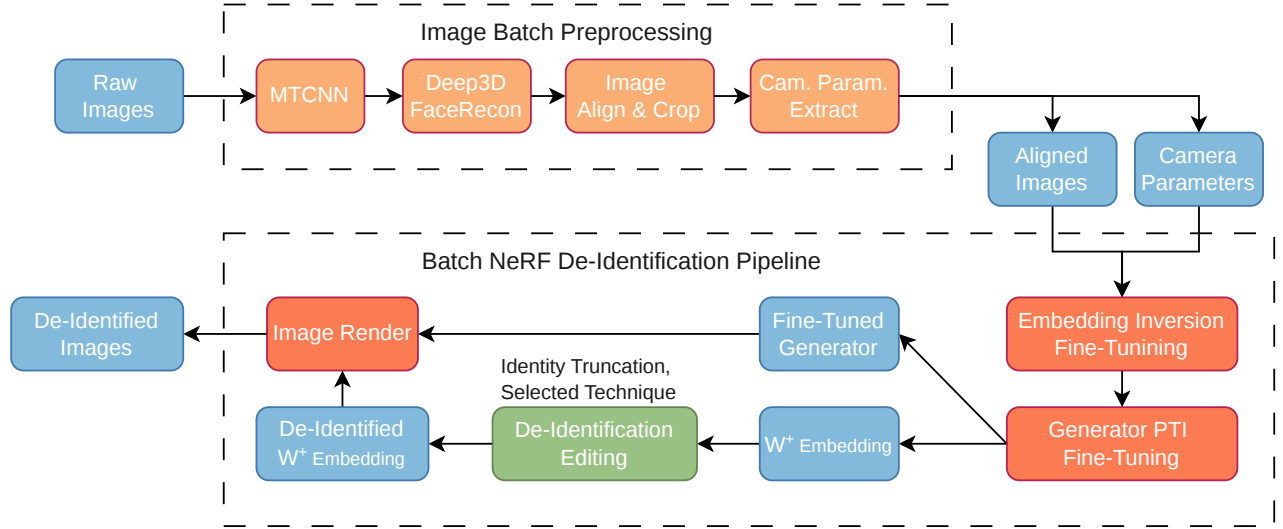


Fig. 2. Overview of the de-identification pipeline. Images are first preprocessed to obtain camera parameters and to produce cropped and aligned face images. During NeRF de-identification, inversion and PTI fine-tuning are applied to optimize the image embedding and fine-tune the generator. De-identification is then achieved by editing the embedding and rendering the resulting image with the fine-tuned generator.

D. Evaluation Experiments

Each image was edited using all three de-identification techniques and rendered at five identity truncation levels:

$$t \in \{0, 0.25, 0.50, 0.75, 1.00\},$$

where $t = 0$ returns the original latent embedding (Full-Identity Render) and higher values apply stronger edits. The images were evaluated using the following metrics:

- **Image Quality.** Each rendered image was compared with its original input image. Reconstruction quality was measured using the Structural Similarity Index Measure (SSIM) and Learned Perceptual Image Patch Similarity (LPIPS). Higher SSIM and lower LPIPS indicate greater visual similarity.
- **Subject Sex.** Sex predictions were obtained for the original and rendered images using the DeepFace classifier [8]. Predictions were compared with the ground-truth labels, and accuracy, precision, recall, and F1 score were reported.
- **Subject Age.** Age predictions were obtained using the DeepFace age regression model [8]. Ground-truth age labels and predicted age were used to compute the Mean Absolute Error (MAE) metric. The standard deviation of the prediction errors and the percentage of predictions within ± 5 years of the ground-truth age were also reported.
- **Identity Recognition.** Genuine and impostor image pairs were constructed using the subject identifiers. AdaFace [9] embeddings were extracted from each image, and pairwise cosine similarity was computed.

Genuine and impostor similarity distributions were presented as histograms. An acceptance threshold τ was selected from the impostor distribution at a False Match Rate (FMR) of 0.001.

The Identity Leakage Rate (ILR) measured the proportion of de-identified genuine pairs that remained accepted as the same identity:

$$\text{ILR} = \frac{\#\{s_{\text{genuine}} \geq \tau\}}{N_{\text{genuine}}}.$$

The De-identification Success Rate (DSR) measured the proportion of genuine pairs that were no longer accepted:

$$\text{DSR} = \frac{\#\{s_{\text{genuine}} < \tau\}}{N_{\text{genuine}}} = 1 - \text{ILR}.$$

The False Identity Match Rate (FIMR) measured the proportion of de-identified impostor pairs were incorrectly accepted as genuine matches:

$$\text{FIMR} = \frac{\#\{s_{\text{impostor}} \geq \tau\}}{N_{\text{impostor}}}.$$

Receiver Operating Characteristic (ROC) curves and the corresponding Area Under the Curve (AUC) were additionally reported for each technique and identity truncation value.

IV. RESULTS

Image quality results are presented in Table I. Across all three de-identification techniques, similarity to the original image decreases as the truncation strength increases. Even at full identity retention, the NeRF renders exhibit a high baseline reconstruction error in both metrics.

TABLE I

MEAN SSIM AND LPIPS IMAGE QUALITY METRICS FOR BASELINE NERF RENDERS AND RENDERS GENERATED USING DIFFERENT DE-IDENTIFICATION TECHNIQUES AND IDENTITY TRUNCATION VALUES.

Technique	Trunc.	SSIM	LPIPS
Full Identity	0.00	0.730	0.337
Middle Average	0.25	0.659	0.389
Middle Average	0.50	0.576	0.450
Middle Average	0.75	0.510	0.501
Middle Average	1.00	0.458	0.540
Random Offset	0.25	0.665	0.393
Random Offset	0.50	0.575	0.466
Random Offset	0.75	0.495	0.534
Random Offset	1.00	0.427	0.594
Image-Matched Residual	0.25	0.602	0.434
Image-Matched Residual	0.50	0.485	0.522
Image-Matched Residual	0.75	0.406	0.584
Image-Matched Residual	1.00	0.352	0.631

Sex classification results are presented in Table II. DeepFace already shows imperfect performance on the original images, and classification metrics decrease further for the full-identity NeRF renders. Across all de-identification techniques, accuracy, precision, recall, and F1 score generally decline as truncation strength increases.

TABLE II

SEX CLASSIFICATION PERFORMANCE OF THE DEEPFACE CLASSIFIER ON ORIGINAL IMAGES, FULL-IDENTITY NERF RENDERS, AND RENDERS GENERATED USING DIFFERENT DE-IDENTIFICATION TECHNIQUES AND IDENTITY TRUNCATION VALUES.

Technique	Trunc.	Accuracy	Precision	Recall	F1
Original	–	0.781	0.845	0.781	0.770
Full Id.	0.00	0.731	0.823	0.731	0.711
Mid. Avg.	0.25	0.702	0.811	0.702	0.674
Mid. Avg.	0.50	0.678	0.804	0.678	0.641
Mid. Avg.	0.75	0.664	0.791	0.664	0.623
Mid. Avg.	1.00	0.657	0.780	0.657	0.615
Rnd. Offset	0.25	0.717	0.815	0.717	0.694
Rnd. Offset	0.50	0.680	0.793	0.680	0.646
Rnd. Offset	0.75	0.649	0.768	0.649	0.606
Rnd. Offset	1.00	0.617	0.712	0.617	0.569
IMR	0.25	0.727	0.814	0.727	0.707
IMR	0.50	0.681	0.727	0.681	0.664
IMR	0.75	0.617	0.629	0.617	0.609
IMR	1.00	0.589	0.593	0.589	0.585

Age prediction results are presented in Table III. The mean absolute error remains similar for the original images and full-identity NeRF renders. As truncation strength increases, all three techniques show higher prediction error and greater error variability. Image-matched residual de-identification produces the largest errors and variance overall.

A. Identity Pair Metric Results

Figure 3 presents the cosine-similarity distributions for full-identity NeRF render pairs. The genuine and impostor

TABLE III

AGE PREDICTION PERFORMANCE OF THE DEEPFACE PREDICTOR ON ORIGINAL IMAGES, FULL-IDENTITY NERF RENDERS, AND RENDERS GENERATED USING DIFFERENT DE-IDENTIFICATION TECHNIQUES AND IDENTITY-TRUNCATION LEVELS.

Technique	Trunc.	MAE	Error STD	Within ± 5 Years (%)
Original	–	6.86	8.44	41.18
Full Id.	0.00	6.92	8.61	42.89
Mid. Avg.	0.25	7.37	9.16	40.08
Mid. Avg.	0.50	8.31	10.27	34.87
Mid. Avg.	0.75	9.02	11.05	32.26
Mid. Avg.	1.00	9.74	11.83	28.76
Rnd. Offset	0.25	7.19	9.00	41.28
Rnd. Offset	0.50	8.13	10.06	36.07
Rnd. Offset	0.75	9.17	11.41	32.57
Rnd. Offset	1.00	10.34	12.82	26.85
IMR	0.25	8.53	10.52	35.57
IMR	0.50	10.41	13.05	30.86
IMR	0.75	12.09	15.03	27.25
IMR	1.00	13.31	16.60	23.95

distributions are clearly separated at the threshold corresponding to an FMR of (0.001). At this threshold, all genuine pairs are correctly accepted as belonging to the same identity.

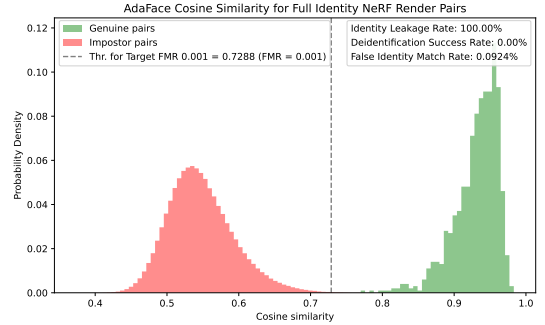


Fig. 3. Baseline full-identity NeRF render cosine similarity distributions for genuine and impostor pairs obtained using AdaFace. The decision thresholds and corresponding verification metrics are reported at an original image FMR of 10^{-3} .

Identity-pair results for Middle Average Interpolation are shown in Figure 4. As truncation increases, the genuine-pair distribution shifts toward and increasingly overlaps with the impostor distribution. At a truncation value of 0.75, over half of the genuine pairs fall below the acceptance threshold. At 1.0, 84.57% of genuine pairs are successfully de-identified. However, the corresponding AUC remains high at 0.9745, indicating that the genuine and impostor scores remain largely distinguishable despite the substantial reduction in genuine-pair acceptance at the fixed threshold.

Figure 5 presents the identity-pair results for Random Offset de-identification. The results follow the same general trend as Middle Average Interpolation but demonstrate stronger de-identification performance. At a truncation

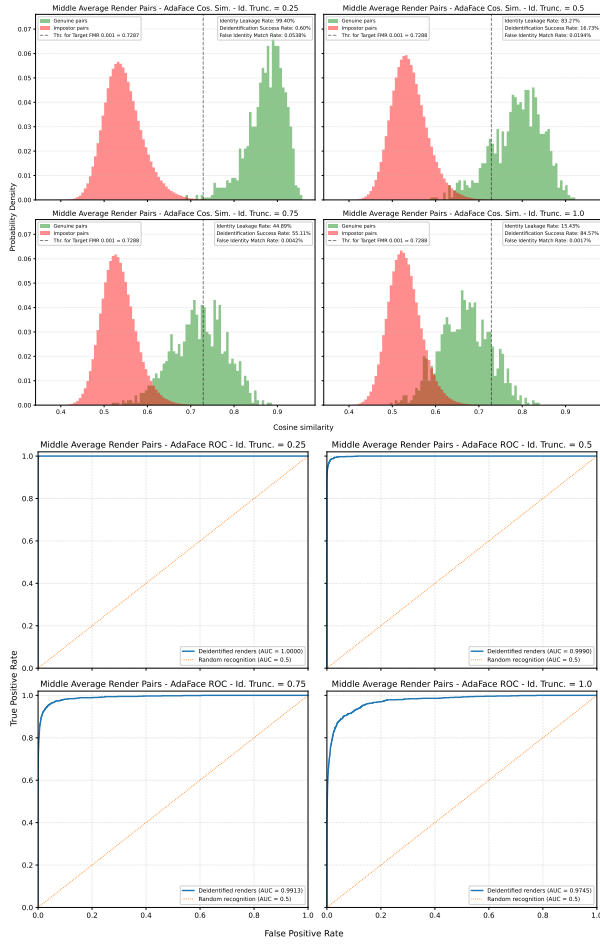


Fig. 4. Middle Average Interpolation cosine-similarity distributions for genuine and impostor pairs (top) and the corresponding ROC curves (bottom), computed with AdaFace across different identity-truncation values.

value of 0.75, Random Offset achieves a DSR of 63.63% with an AUC of 0.9853. At 1.0, the DSR increases to 94.99%, while the AUC decreases to 0.9403. These results indicate that Random Offset produces more aggressive identity modifications than Middle Average Interpolation.

Figure 6 presents the identity-pair results for Image-Matched Residual de-identification. This technique achieves the strongest de-identification performance among the evaluated methods. At a truncation value of 0.5, it achieves a DSR of 61.02% and an AUC of 0.9761. At 1.0, the DSR increases to 97.29%, while the AUC decreases to 0.8466. The lower AUC indicates reduced separability between genuine and impostor pairs, suggesting that identity recognition becomes substantially less reliable after de-identification.

B. Qualitative Results

Figure 7 presents representative renders from each de-identification technique across all identity-truncation levels.

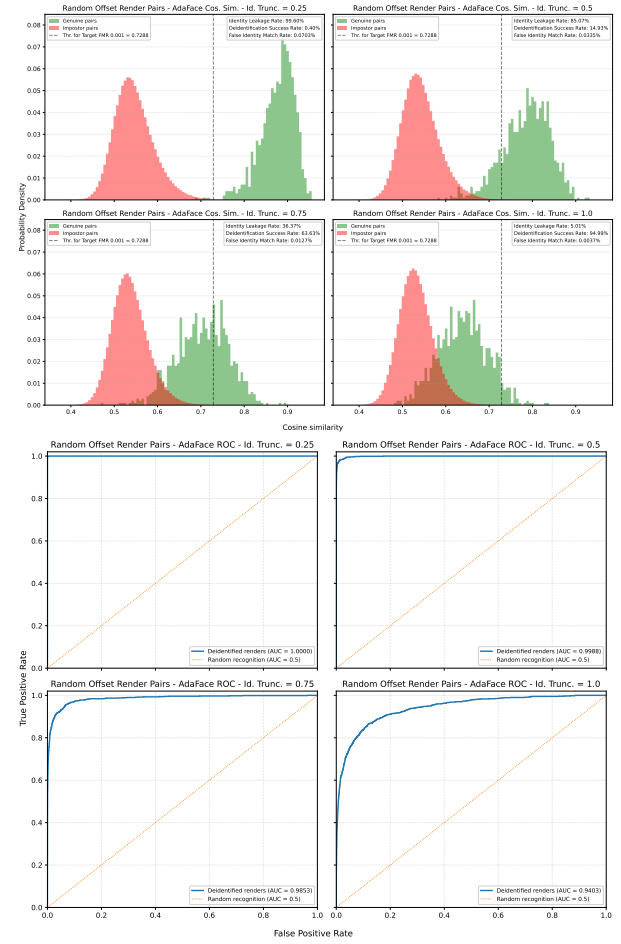


Fig. 5. Random Offset render cosine-similarity distributions for genuine and impostor pairs (top) and the corresponding ROC curves (bottom), computed with AdaFace across different identity-truncation values.

The full-identity renders at truncation 0.00 exhibit reconstruction artifacts, including backgrounds and crops bleeding into facial regions and the loss of high-frequency details.

Random Offset produces the most visually destructive edits. Because the random latent directions have no semantic interpretation, increasing truncation frequently introduces severe facial distortions and inconsistent lighting.

Middle Average Interpolation generally produces plausible faces and converges toward a predictable appearance, often characterized by a relaxed smile, while retaining some features of the original subject.

Image-Matched Residual renders highlight limitations in age preservation, consistent with the results in Table III. At truncation 1.00, the apparent ages of the generated subjects often differ substantially from that of the original image.

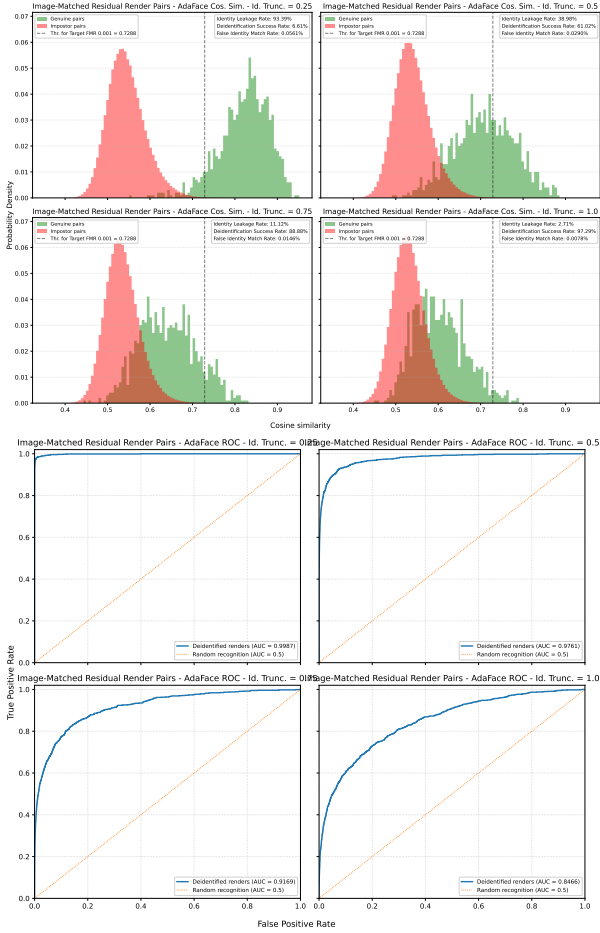


Fig. 6. Image-Matched Residual render cosine-similarity distributions for genuine and impostor pairs (top) and the corresponding ROC curves (bottom), computed with AdaFace across different identity-truncation values.

V. DISCUSSION

The MORPH-II dataset introduces several limitations that may have influenced the reported results. Although it contains age and sex labels, it remains racially imbalanced, with African-American subjects forming the majority of samples. A more balanced evaluation would require explicit race labels so that the dataset could be stratified across demographic groups.

Dataset image quality may also have limited the reconstruction performance. Samples are low-resolution, tightly cropped, contain black borders, or exclude parts of the face. These properties may contribute to the degradation observed even in full-identity renders, particularly in the image quality, sex classification, and age prediction results. Higher-resolution images with complete facial regions and cleaner backgrounds may produce more faithful renders.

The evaluated de-identification techniques operate without explicit semantic control. Their latent edits are not conditioned on attributes such as age, sex, or race, which can

lead to visual artifacts and changes to these characteristics. This limitation is reflected in the reduced sex classification performance, increased age prediction error, and visible variation across strongly de-identified renders.

Future work should therefore focus on semantic latent editing. A trained or fine-tuned embedding editor could be used to modify identity information while preserving selected demographic and visual attributes. Such an approach may provide a better balance between de-identification strength, image quality, and attribute retention.

VI. CONCLUSION

NeRF-based facial reconstruction shows strong potential for face de-identification. Its ability to preserve three-dimensional structure and produce viewpoint-consistent renders makes NeRFs suitable for privacy-preserving face synthesis. The evaluated latent-space techniques substantially reduced identity-recognition performance, although stronger de-identification degraded image quality and altered biometric attributes. Future research should focus on semantically controlled latent editing that removes identity while preserving selected non-identity characteristics.

REFERENCES

- [1] B. Meden, P. Rot, P. Terh rst, N. Damer, A. Kuijper, W. J. Scheirer, A. Ross, P. Peer, and V.  truc, "Privacy-enhancing face biometrics: A comprehensive survey," *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 4147–4183, 2021.
- [2] B. Meden,  . Emer i , V.  truc, and P. Peer, "k-same-net: k-anonymity with generative deep neural networks for face deidentification," *Entropy*, vol. 20, no. 1, p. 60, 2018.
- [3] H. Zhang, T. DAI, Y. Xu, Y.-W. Tai, and C.-K. Tang, "Faced-nerf: Semantics-driven face reconstruction, prompt editing and relighting with diffusion models," *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [4] A. W. Rawls and K. Ricanek, "Morph: Development and optimization of a longitudinal age progression database," in *Biometric ID Management and Multimodal Communication*, J. Fierrez, J. Ortega-Garcia, A. Esposito, A. Drygajlo, and M. Faundez-Zanuy, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 17–24.
- [5] C. S. Panuganti, "MORPH-2," Kaggle dataset, n.d., accessed: 2026-07-28. [Online]. Available: <https://www.kaggle.com/datasets/chiragsaipanuganti/morph>
- [6] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," *IEEE Signal Processing Letters*, vol. 23, no. 10, p. 1499–1503, Oct. 2016. [Online]. Available: <http://dx.doi.org/10.1109/LSP.2016.2603342>
- [7] Y. Deng, J. Yang, S. Xu, D. Chen, Y. Jia, and X. Tong, "Accurate 3d face reconstruction with weakly-supervised learning: From single image to image set," in *IEEE Computer Vision and Pattern Recognition Workshops*, 2019.
- [8] Y. Taigman, M. Yang, M. Ranzato, and L. Wolf, "Deepface: Closing the gap to human-level performance in face verification," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 1701–1708.
- [9] M. Kim, A. K. Jain, and X. Liu, "Adaface: Quality adaptive margin for face recognition," 2023. [Online]. Available: <https://arxiv.org/abs/2204.00964>

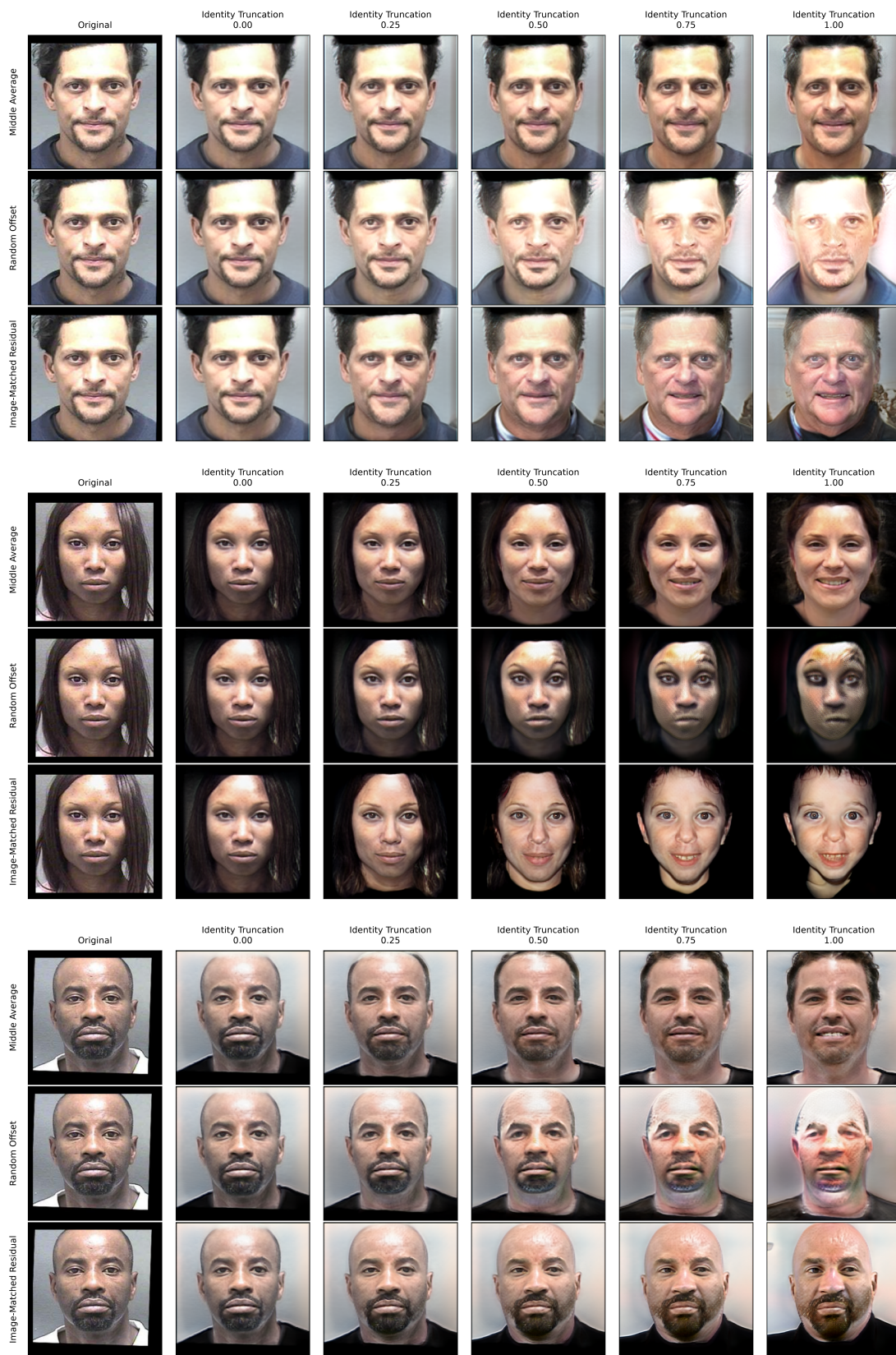


Fig. 7. Examples of original images and corresponding NeRF renders generated using different de-identification techniques across increasing identity-truncation levels.